
Real-Time User Behavior Anomaly Detection Platform

Checkpoint 1 — Foundation & Use Case Analysis

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Project Overview

A production-oriented real-time anomaly detection system designed for a **banking environment**. The platform ingests user events from multiple sources, processes them through a scalable streaming architecture, extracts behavioral features, and applies machine learning models to detect deviations from normal user patterns in near real time.

Design philosophy: the system must be auditable, explainable, resilient to failure, and sensitive enough to catch real threats without drowning analysts in false positives.

Detection Scenario Catalog

Scenario 1 — Geo-Velocity Anomaly

Signal: Login from a geographically impossible location given the user's last known position and elapsed time.

Example: User logs in from Tunis, then 2 minutes later a login appears from Brazil.

Response: **Step-up authentication** — confirm identity with the user before allowing further actions.

Scenario 2 — Suspicious Large Deposit (AML)

Signal: Sudden large inflow disproportionate to the account's historical balance.

Example: Account with a typical balance of \$1,200 receives \$50,000.

Response: **Flag for compliance review** — do not block, but route to AML team for source-of-funds analysis.

Scenario 3 — Fraudulent Transaction Pattern

Signal: Transaction that deviates significantly from the user's established behavioral profile across multiple dimensions (amount, recipient, timing, device, location).

Response: **Block transaction** — hold funds and alert fraud team immediately.

Scenario 4 — Card Testing

Signal: Rapid sequence of micro-transactions across different merchants in a short window.

Example: 20 transactions of 0.50 in 3 minutes.

Response: **Block card** — likely stolen card validation attempt.

Scenario 5 — Account Takeover Sequence

Signal: Rapid succession of sensitive account changes: password change → email change → new device login → funds transfer.

Key insight: Each action alone is normal; the *sequence within a compressed time window* is the anomaly.

Response: **Lock account** — require identity verification to restore access.

Scenario 6 — Dormant Account Activation

Signal: Account inactive for months suddenly shows daily transaction activity.

Response: **Flag for review** — possible compromised account used as a mule.

User Behavior Profile Design

Core Principle

The system does **not** analyze raw event logs in real time. Instead, it maintains a **pre-computed behavioral profile** per user that summarizes their “normal” and can be compared against incoming events in milliseconds.

Critical distinction: profiles store **distributions and sets**, not single values. An average transaction amount is insufficient; the system needs spread (standard deviation, percentiles) to judge how unusual a new event is *for that specific user*.

Profile Fields

Privacy Principle

The model never sees actual credentials, emails, or personal content. It only sees **behavioral metadata**: timestamps, frequencies, counts, and categorical flags. Sensitive fields are either hashed or reduced to event indicators (e.g., “password changed = true” rather than the password itself).

Profile Evolution Strategy

Profiles must evolve over time to reflect legitimate behavioral changes (moving cities, salary increase, new business), but uncontrolled updates introduce the risk of **profile poisoning**: an attacker gradually shifts the profile to normalize malicious behavior before striking.

Field	Type	Description
account_id	Identifier	Unique user identifier
known_locations	Set + frequency	Set of locations with login frequency per location
known_devices	Set + frequency	Registered devices with last-seen timestamps
amount_distribution	Distribution	Mean, std dev, percentiles of transaction amounts
active_hours	Distribution	Histogram of activity across hours of day and days of week
tx_frequency	Distribution	Typical number of transactions per day/hour; inter-transaction gap
known_recipients	Set + frequency	Regular payees and organizations with transfer frequency
last_password_change	Timestamp + rate	Date of last change and historical change frequency
last_email_change	Timestamp + rate	Date of last change and historical change frequency
account_age	Duration	Time since account creation
dormancy_indicator	Duration	Time since last activity

Table 1: User behavior profile schema (v1)

Mitigation Strategies

1. **Decay Windows** — Maintain both a short-term profile (e.g., 30 days) and a long-term baseline (e.g., 6 months). Compare incoming events against both.
2. **Verified-Only Updates** — Only update the profile from activity that passed authentication and was not flagged as suspicious. Flagged events are quarantined until cleared.
3. **Rate-Limited Drift** — Cap how much any profile dimension can shift per time window. Sudden jumps in the profile are rejected even if individual events were not flagged.

Key Design Insights So Far

1. Individual events may look normal; **sequences and context** reveal anomalies (especially account takeover).
2. High sensitivity without precision creates a **false positive flood** that buries real threats — detection must be tuned per scenario.
3. Different anomalies require different responses: **block**, **flag for review**, or **step-up auth**.
4. The system needs **real-time speed** with **historical context** — solved by pre-computed profiles.
5. Privacy by design: behavioral metadata only, never raw secrets.

Next Steps

Define the system architecture: trace the journey of a single event from ingestion to decision.

Identify core components: event ingestion, feature extraction, profile store, inference engine, alert routing.

Choose technologies for each component.